Highlights

Reliability of Short Molecular Dynamics Runs as Screening Labels: Noise Ceilings, Time Dependence and Adaptive Replicate Allocation

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- Short-run labels drift with trajectory age as well as observation lag
- Replicas quantify reliability and a model-dependent prediction ceiling
- Adaptive replication saves computation in calibrated screening simulations
- Allocation targets expected short-run labels, not long-time transport
- Code and compact derived inputs reproduce the analysis without raw trajectories

Reliability of Short Molecular Dynamics Runs as Screening Labels: Noise Ceilings, Time Dependence and Adaptive Replicate Allocation

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Abstract

Molecular dynamics (MD) supplies noisy labels whose values depend on simulation time, for materials screening and machine learning. We audit these labels in HTP-MD (6,137 runs of 5 ns and 133 of 50 ns), separating dependence on observation lag from changes with trajectory age. The first 5 ns segment is a median 1.63 times faster than segments at 25–50 ns; concurrent energy and polymer-mobility changes support incomplete relaxation. From 94 replicated screening systems, one end-point label has reliability 0.44 (95% CI 0.27 to 0.61), implying a population correlation ceiling of 0.66 under an additive measurement model. A fingerprint random forest reaches 0.69; the uncertainty of the ceiling precludes a claim of model saturation. Comparisons with same-structure runs attribute about half of the run-to-run variance to structure under the assumed decomposition, without establishing that this contribution persists indefinitely. In simulated campaigns calibrated with these data, adaptive replication matches four uniform replicates using 1.9 runs per candidate for the top 10% and 1.3 for the top 1% by expected 5 ns label, not by long-time transport. A real-run check cannot confirm the predicted gain. Analyses of anion diffusivity and cluster conductivity extend the audit within the same dataset.

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Keywords: molecular dynamics, simulation output analysis, uncertainty quantification, label noise, simulation budget allocation, high-throughput screening, polymer electrolytes

1. Introduction

Molecular dynamics (MD) increasingly supplies labels for thousands of materials, to rank candidates or train machine-learning models [1]. Polymer electrolytes illustrate the challenge: ion transport couples to slow polymer dynamics [2, 3], while high-throughput campaigns can afford only short trajectories. HTP-MD distributes thousands of 5 ns runs and a smaller set of 50 ns runs, including raw trajectories and derived properties [1, 4].

The dataset's authors already quantified variability between amorphous initializations, compared predictions against replicated simulations and used multi-task learning to correct differences between short and long runs [1]. Finite-trajectory diffusion estimation [5–8], MD uncertainty analysis [9, 10] and noisy-label learning [11] provide established tools. Replicate variability also bounds population prediction metrics [12–14]. In stochastic simulation, incomplete warm-up, independent replications and selection under a budget are classical output-analysis problems [15–19].

The question addressed here is how these tools, used together, change the interpretation and use of labels from a real screening campaign. We make three contributions. First, we distinguish lag dependence of apparent diffusivity from nonstationarity with trajectory age, by comparing consecutive segments of equal duration. Second, we express replica variability relative to the screening population, estimate its implications for prediction and selection, and compare a simple predictor against the uncertain ceiling. Third, we evaluate known replication strategies in campaigns calibrated to the measured noise, with a limited check on real runs.

The central target is the expected label of the short-run protocol across initializations, which can differ systematically from long-time transport. This distinction separates the precision improved by replication from the time-dependent bias it does not necessarily remove. The result is a reproducible audit connecting trajectory nonstationarity, population-level label reliability and the performance of replication policies under a specified screening objective.

2. Methods

2.1. Data set

We use every trajectory listed by the public HTP-MD web application [4] on 6 to 9 October 2026: 6,270 trajectories of amorphous polymer/LiTFSI systems simulated with the PCFF+ force field at 353 K, 6,137 for 5 ns and 133 for 50 ns (Table S1). For each trajectory the platform distributes the MSD curves of Li^+ and of the anion at 100 lags, derived transport properties and the raw trajectory, with frames every 2 ps. The storage paths identify each system. Every long run has a 5 ns run of the same system (96 screening polymers and 37 taken from the literature), and, as in the original study [1], the two runs of such a pair start from the same initial amorphous structure. In addition, 94 screening systems were simulated twice for 5 ns from different initial structures. The storage paths label these second runs as a random sample of 100 screening systems (`rand_100`), and 92 of the 96 long screening runs belong to the same systems. Neither subset is selected by mobility: the short-run labels of the replicated systems and of the systems with a long run are distributed as those of the rest of the screening set (Kolmogorov-Smirnov $p = 0.61$ and 0.65; Levene $p = 0.82$ and 0.65), supporting their use to estimate population-average noise under the documented sampling design. A nonsignificant distribution test does not establish representativeness of noise across chemical families; the selection and availability audit is reported in the Supplementary Material. Only the 37 literature polymers were chosen deliberately, and they are analysed separately. Every system contains 50 LiTFSI ion pairs at a nominal molality of 1.5 mol kg^{-1}, equilibrated for 5 ns before production [1]; from the starting structures, the systems have 2,328 to 7,541 atoms (median 6,002) in 1 to 15 polymer chains (median 9), in cubic boxes of 3.0 to 4.2 nm side (ion and chain counts verified on 125 complete structure files).

2.2. Labels and their time dependence

For an MSD curve averaged over time origins and ions, $\mathrm{MSD}(\tau) = \langle|\mathbf{r}_i(t+\tau) - \mathbf{r}_i(t)|^2\rangle_{i,t}$, we define the apparent diffusivity $D_{\mathrm{app}}(\tau) = \mathrm{MSD}(\tau)/6\tau$. The diffusivity distributed by the platform is the *end-point* estimate $D_{\mathrm{end}}(T) = \langle|\mathbf{r}_i(T) - \mathbf{r}_i(0)|^2\rangle_i/6T$, which uses only the first and last frames [20]; we also use the origin-averaged $D_{\mathrm{app}}(T/2)$. Diffusion corresponds to a lag-independent D_{app} and an MSD exponent $\beta = \mathrm{d}\ln\mathrm{MSD}/\mathrm{d}\ln\tau = 1$. We estimate an effective exponent by a least-squares fit of $\ln\mathrm{MSD}$ against $\ln\tau$

over lags between 20% and 60% of the run, and call a run diffusive when $\beta \geq 0.9$; this is an operational classification, and its sensitivity to the window is reported in Table S4. For the long runs we compute $D_{\mathrm{app}}(\tau)$ from 0.5 ns to 25 ns and report its ratio to the reference $D_{\mathrm{app}}(25\,\mathrm{ns})$ and the Spearman correlation of the rankings. The reference is a finite-lag quantity, not a converged coefficient. To test whether the beginning of a run differs from its later part, we split each long run into ten consecutive 5 ns segments and compute, from the data of each segment alone, the end-point and origin-averaged estimates and the effective exponent; each segment is compared with the geometric mean of the segments between 25 and 50 ns. In the same segments we follow the mobility of the anion and of polymer atoms, the potential energy and pressure (production is at constant volume), and the lithium-anion clusters computed during the simulations (Supplementary Material).

2.3. Noise model, reliability and noise ceiling

All noise statistics are computed on $\ln D$. Following Ref. [1], we model each label as a system-level value plus run-specific noise. Here, the system-level value is the expected 5 ns label across initial structures and dynamical realizations. It can include biases shared by short runs, such as the early slowdown reported below; it is not the long-time transport coefficient. Thus, the reliability and noise ceiling describe how well short-run labels can be predicted. A separate, system-dependent bias distinguishes those labels from the long-time property. For the replicated systems, the standard deviation of the difference between the two runs divided by $\sqrt{2}$ estimates the noise σ_{run} of one run. With σ_{tot} the spread of one original short-run label per screening system (literature polymers excluded), the reliability of one run is

$$r = 1 - \sigma^2_{\mathrm{run}}/\sigma^2_{\mathrm{tot}}, \tag{1}$$

the fraction of the variance between candidates that reflects differences between systems. We also report the one-way intraclass correlation computed within the replicated subset. For a predictor independent of the noise of an evaluation run, the additive model implies a population correlation ceiling $\sqrt{r}$ and a population explained-variance ceiling r, the bound used for experimental data [12, 13] and known as the noise ceiling in computational neuroscience [14]; the mean of n runs has reliability $nr/[1+(n-1)r]$ [21, 22]. To translate reliability into selection we simulate Gaussian populations with the measured variances and record the fraction of the truly best 5%, 10% and 20% recovered when selecting by the mean of n runs.

Two sources of noise are separated under an additive model with independent structural and dynamic contributions, so the split is a statistical attribution conditional on that model rather than a causal decomposition. Replicas start from different structures, so their difference contains structural and dynamic noise. The 5 ns run of a pair and the first 5 ns of its long run start from the same structure, so their difference estimates dynamic noise, σ_{dyn}, if the two dynamical realizations are independent; the two are distinct dynamical realisations (their labels are not identical), although the platform does not document whether they share any part of the equilibration. Positive correlation from a shared history would underestimate dynamic noise and overattribute variance to structure. Then $\sigma^2_{\mathrm{str}} = \sigma^2_{\mathrm{run}} - \sigma^2_{\mathrm{dyn}}$, with σ_{run} from the 94 replicated systems and σ_{dyn} from the 96 screening pairs. The short run of a pair is also the original replica of that system, so the two differences share a run; we therefore repeat the split on the 92 systems that have both, resampling them jointly. We test whether the noise depends on mobility by the rank correlation between the absolute difference of two replicas and their mean. Intervals throughout are obtained by bootstrap over systems (2000 resamples).

2.4. Machine-learning demonstration

To compare a real model with the ceiling we predict the end-point $\ln D$ of the original 5 ns run of each system from its monomer alone. The input is the Morgan count fingerprint [23] (radius 2, 2048 bits; RDKit [24]) of the monomer SMILES distributed by HTP-MD, including the Au/Cu connector markers that mark the connection points, so chain architecture is not represented. The model is a random forest (scikit-learn [25]; 400 trees, a fraction 0.3 of the features per split, at least two samples per leaf), with these settings fixed in advance and not tuned. It is trained on the 6,003 systems with an original screening run, excluding literature polymers and the 1 system available only as a replica, and evaluated by 5-fold cross-validation split by system, so that every prediction is out of sample. As a stricter split we group the monomers into chemical families by Butina clustering [26] of the Tanimoto distances of their binary Morgan fingerprints, at two similarity thresholds, and keep each family within one fold. Intervals for the correlations are obtained by bootstrap over systems of the out-of-fold predictions. For the literature polymers we fit all screening systems and evaluate externally, without using literature labels for training. To measure the effect of splitting by trajectory instead of by system, we repeat the cross-validation

with every trajectory, including the second runs of the replicated systems, as its own group, and compare the two splits on the replicated pairs whose two runs fell in different folds (77 of the 94; in the other 17 both runs were in the same test fold).

2.5. Selection from paired runs

For the 133 pairs we compare the end-point label of the 5 ns run with the reference of the long run of the same system: median ratio, Spearman correlation (separately for screening and literature polymers) and the recall of the best 5%, 10% and 20% systems, that is, the fraction of the k best according to the reference that are also among the k best according to the short run, against the recall k/n of a random choice. Because the two runs share their initial structure, this comparison does not sample independent initial structures and can overstate agreement.

2.6. Truncated trajectories and stopping rules

From the raw trajectory of each long run we extract the unwrapped lithium positions, subtract the centre-of-mass motion of the whole system as the released library does [20] (validation in the Supplementary Material), truncate the trajectory to its first T nanoseconds and estimate the diffusivity from that prefix alone with four estimators: the end-point, the origin-averaged $D_{\mathrm{app}}(T/2)$, one sixth of the slope of a linear fit to the MSD over $0.2T$ to $0.5T$, the conventional procedure [10], and a power law $\mathrm{MSD} = A\tau^{\beta}$ fitted over $0.1T$ to $0.5T$ and extrapolated to τ_{ref}, $\widehat{D_{\mathrm{app}}} = A\tau_{\mathrm{ref}}^{\beta-1}/6$. Estimates are compared with the reference of the full run through the typical multiplicative error $\exp(\mathrm{median}\,|\ln(\hat{D}/D_{\mathrm{ref}})|)$; because the reference shares data with the prefix, we repeat the comparison against references at other lags and against the late slope of the MSD. Stopping rules stop a run when its effective exponent reaches a threshold or when the bootstrap coefficient of variation of an estimate falls below a tolerance; the bootstrap resamples ions and does not model correlations between their motions, so its precision is conditional; runs that never stop are simulated to 50 ns, and cost and error are averaged over all runs. Thresholds are chosen on a random half of the systems and evaluated on the other half (500 partitions).

2.7. Adaptive allocation of replicates

Consider a campaign of $N = 6{,}003$ candidates, a hypothetical population the size of the original screening set, with a budget of b 5 ns runs per candidate

on average, whose goal is the top k, a ranking-and-selection problem [18]. The true top set is defined by expected 5 ns labels across initializations, not by long-time diffusivity. System values are drawn from $\mathcal{N}(0, \sigma^2_{\mathrm{sys}})$ with $\sigma^2_{\mathrm{sys}} = r\sigma^2_{\mathrm{run}}/(1-r)$, and each run adds noise whose standard deviation varies linearly with mobility so as to reproduce the measured noise of the slow and fast halves; every replicate starts from a new structure and carries the full noise. The unit of cost is therefore a complete new-structure run, which in the HTP-MD protocol comprises 5 ns of equilibration and 5 ns of production; budgets and savings below are counted in such runs. A normal prior and a Gaussian approximation of the posterior, with the noise evaluated at the observed mean of each candidate rather than at its unknown value, give the mean and uncertainty of each candidate. We compare three allocations of the budget. *Uniform*: b runs for every candidate. *Successive halving* [27]: one run for all, then further runs for the best $4k$, $2k$ and k, in rounds that restart from $4k$ until the budget is spent. *Boundary-focused*: one run for all, then batches of 5% of N further runs to the candidates with the largest $p(1-p)$, where p is the posterior probability of lying above the current top-k cut-off (the k-th largest posterior mean, treated as fixed within a batch), in the spirit of optimal computing budget allocation [19] and of its version for selecting a subset of the best designs [28]. The final selection is the top k by posterior mean, and the metric is the recall of the true top k, averaged over 200 simulated campaigns (Monte Carlo standard error at most 0.4 percentage points for these three policies). As a simple reference we add a *two-stage re-screening*: one run for all, then the rest of the budget spread evenly over a fixed shortlist of the best $2k$, $4k$ or $8k$ after the first run, taking for each budget the best of the three shortlists. Equivalent budgets are approximated by linear interpolation of the Monte Carlo mean recall curves after taking a nondecreasing envelope. Robustness is checked with the empirical shape of the short-run distribution instead of a normal one, with the origin-averaged label, with a constant instead of a mobility-dependent noise, with a misestimated reliability (the runs follow the measured noise, but the allocation and the selection use the ends of its 95% interval), and with the true reliability at either end of that interval. The algorithm and its parameters are given in the Supplementary Material.

The public data allow a partial check on real runs. In the replicated systems, the first run is the original screening run and the second starts from a new structure, and 92 of them also have a long run, which starts from the same structure as the first. We use them in two ways, First, with noise

parameters estimated leaving out the system being evaluated, from one run of each system the model predicts the probability that the other run exceeds the top-k cut-off; we compare these predictions with what happens (Brier score against the base rate and calibration slope). Second, we run a small real campaign: every system receives its first run, and 46 second runs go either to the candidates with the largest $p(1-p)$ or to randomly chosen ones; the selection is evaluated against the long runs, which are independent of the second runs. The campaign noise parameters are fitted to the available campaign pairs, so this is an exploratory check rather than independent validation. Its bootstrap refits the noise and reruns the complete allocation at a fixed budget; the random control is averaged over 50 placements per resample.

3. Results and Discussion

3.1. Labels depend on the simulated time

After 5 ns the median effective exponent of the lithium MSD is 0.43 (10th to 90th percentile range 0.26 to 0.60): 97.8% of the runs have $\beta < 0.7$ and 2 of 6,137 meet the diffusive criterion (Fig. 1). Ten times more simulation moves the distribution but does not resolve the problem: at 50 ns the median is 0.65 and 5 of 133 runs are diffusive (0 to 19 depending on the fitting window, Table S4). In the long runs the apparent diffusivity falls steadily with the lag (Fig. 1): at 5 ns it is a median 1.95 times its value at 25 ns (10th to 90th percentile range 1.47 to 2.40), and only 73.7% of the systems are within 25% of the reference at 15 ns (Table S6). Rankings are far more stable: the Spearman correlation with the ranking at 25 ns is 0.91 already at 0.5 ns and 0.96 at 5 ns. A short-run label is therefore an apparent, window-dependent quantity whose value should not be compared across studies with different lengths, but whose ordering carries information.

Labels also depend on *when* the time is simulated (Fig. 2). Analysed with its own data only, the first 5 ns segment of a long run is a median 1.63 times faster than the segments between 25 and 50 ns (95% interval 1.52 to 1.81; 1.47 with the origin-averaged estimator), and it is faster in 93% of the systems (Wilcoxon signed-rank $p = 3.6 \times 10^{-22}$). The slowdown is not a short transient: the apparent diffusivity keeps falling by a factor of 0.92 every 10 ns between 5 and 50 ns, in 85% of the systems, and the effective exponent of the segments decreases from 0.48 in the first to 0.31 in the late ones. Lithium mobility therefore keeps decreasing throughout the runs,

despite the 5 ns equilibration that precedes production [1], and the whole system is still relaxing (Table S7). At constant volume, the potential energy falls between the first and the late segments in all 133 systems, by a median 0.014 kcal mol^{-1} per atom, 6.6 times the fluctuation of its 100 ps averages, and the anion and the polymer slow down with lithium ($\times$1.46 and $\times$1.43). Across systems, the slowdown of lithium follows that of the polymer (Spearman 0.68, 95% interval 0.57 to 0.76) and the relaxation of the energy (0.58), but not the change in ion clustering, which is small and uncorrelated with it ($-$0.03). The early, faster window is thus consistent with a polymer that has not finished relaxing after the equilibration, as expected if lithium transport is slaved to segmental dynamics [2, 3]; these are correlations, not a demonstration of cause. In simulation terms this is an initialization bias that the fixed equilibration does not remove. Detecting the end of equilibration from the trajectory itself [29], as truncation heuristics do for the warm-up of other stochastic simulations [15, 16], would be a natural way to start production per system. The effect is stronger for the screening polymers (1.80) than for those taken from the literature (1.28).

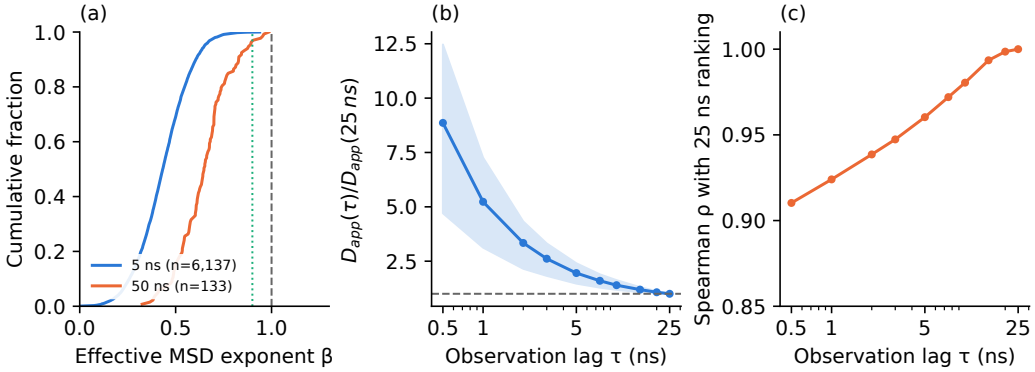


Figure 1: Lag dependence of lithium mobility. Left: effective MSD exponents in short and long runs, with the operational threshold 0.9 and diffusive exponent 1 marked separately. Middle: $D_{\mathrm{app}}(\tau)/D_{\mathrm{app}}(25\,\mathrm{ns})$ in the long runs (median and 10th–90th percentiles). Right: ranking correlation with the 25 ns reference. Lag comparisons share trajectory data.

3.2. Run-to-run noise and its sources

A single 5 ns run deviates from the value of its system by a typical factor of 1.43 for the end-point label ($\sigma_{\mathrm{run}} = 0.35$ in log units) and 1.32 for the origin-averaged one, estimated from pairs of runs of the same system from

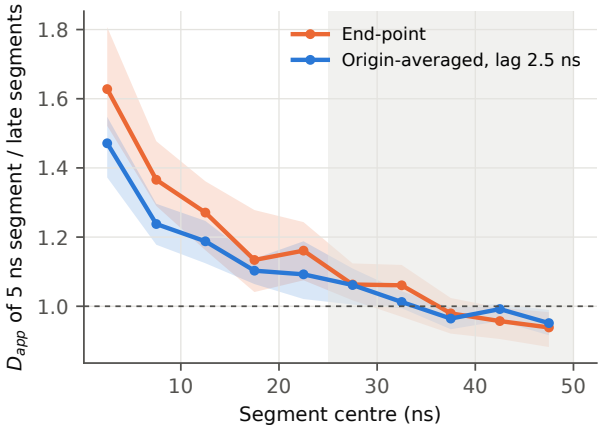


Figure 2: Apparent diffusivity of consecutive 5 ns segments of the 133 long runs, each computed from its own data, relative to the segments between 25 and 50 ns (shaded): median over systems and 95% bootstrap interval.

different structures (Table S8). Against the spread of the original screening population, the reliability of one end-point label is 0.44 (95% interval 0.27 to 0.61), and 0.54 (0.40 to 0.67) for the origin-averaged label; the intraclass correlation within the replicated subset agrees (0.53 and 0.62). Roughly half of the variance between screened candidates is therefore noise.

The noise has two comparable sources. The 5 ns run of each pair agrees with the first 5 ns of its long run, which starts from the same structure, to a median ratio of 1.00 with a noise of 0.24, against 0.35 between replicas from different structures: the initial structure accounts for 55% (28 to 73%) of the run-to-run variance and the dynamics for the rest; on the 92 systems with both comparisons, resampled jointly, the share is 57% (44 to 69%). A new pilot of 49 runs in 10 systems gives a compatible same-structure dispersion (0.23 in log units; Section S6). This protocol check does not validate adaptive allocation or resolve the structural share. The originals, the replicas and the long runs were produced in different batches of the data set, but none is shifted or differently spread (Supplementary Material, after Table S8). Resampling the structure and averaging the dynamics both reduce the noise. A replicate that starts from a new structure samples both sources, as ensembles of independent runs do in other MD applications [30], but it also costs a new equilibration; this is the classical choice between independent replications and one longer run [17]. Under this decomposition, removing dynamic noise

while keeping structural memory unchanged would give a reliability of 0.58, compared with 0.61 for the mean of two independent structures. This is a conditional calculation, not an upper bound for arbitrarily long MD: these data cannot establish how quickly structural memory fades, and extension changes the observation window as well as its precision. The noise also depends on mobility: it is 0.40 in the slower half of the replicated systems and 0.30 in the faster half (rank correlation between the absolute difference and the level -0.26, $p = 0.01$), whereas the learning model of the original study assumes noise independent of the polymer [1]. Two candidates measured once each differ by a factor of 1.65 from noise alone at one standard deviation and up to 2.67 at the 95% level.

Ion aggregation provides limited chemical context. Systems with more lithium in large clusters are slower and their short labels are more overestimated, but associations with bias and ranking error weaken after controlling for long-run mobility and their intervals include zero (Supplementary Material). These exploratory associations do not identify a rearrangement mechanism. Neither aggregation descriptor has a clear association with replica noise; with two runs per system this does not rule out chemical differences in noise.

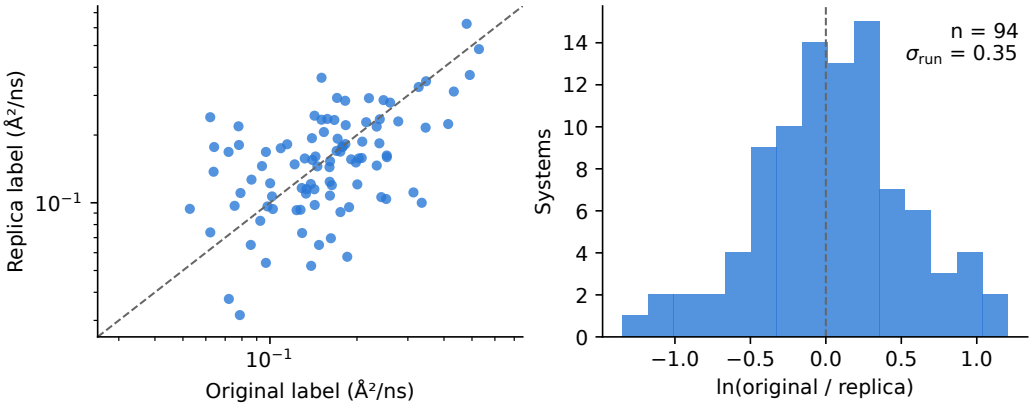


Figure 3: Direct evidence of end-point label variability in the 94 replicated screening systems. Left: original versus independent-initialization label (log axes; dashed identity). Right: differences of natural-log labels; the noise estimate is their standard deviation divided by $\sqrt{2}$. Differences include both structural and dynamic variation.

3.3. A noise ceiling for learned predictors

Under the additive measurement model, the reliability implies a population correlation ceiling of 0.66 (95% interval 0.52 to 0.78) against an independent end-point label, and an explained-variance ceiling of 0.44 (Table 1). For the origin-averaged label these are 0.74 and 0.54. A finite-sample correlation can exceed the estimated ceiling. The selection recalls in the table additionally assume Gaussian system values and noise. Averaging independent initializations raises reliability: two end-point runs give 0.61, four give 0.76, and reaching 0.8 requires 6 runs (4 for the origin-averaged label).

A fingerprint random forest gives out-of-fold Pearson correlation 0.69 (95% interval 0.68 to 0.71; $R^2 = 0.48$) on the same screening population (Table 2). This is compatible with the estimated ceiling, but the wide ceiling interval prevents a claim that model performance has saturated. Chemical-family splits give 0.69 and 0.69; their nearest training neighbours still have appreciable similarity (Supplementary Material), so they test interpolation within this chemical space.

In replicated systems, correlation rises from 0.64 and 0.68 against the single labels to 0.76 against their mean, consistent with the measurement-model prediction 0.78 (gain 95% interval 0.07 to 0.13). Replicated evaluation labels therefore help distinguish model error from evaluation noise. They do not make better models unnecessary. Xie et al. already demonstrated denoising and estimated performance equivalent to about four simulations for conductivity [1]; our analysis expresses the label uncertainty relative to the population and its consequences for metrics.

Against long-run screening references, the model and a system's own short run have rank correlations 0.68 and 0.68. For literature polymers evaluated externally, these are 0.44 and 0.93. Shared initial structures favour the short-run comparison, and neither reference establishes equilibrium transport. System-level splits remain necessary; the trajectory-split check is reported in the Supplementary Material.

3.4. Selecting candidates from one short run

For the 133 systems simulated at both lengths, the end-point label of the 5 ns run is a median 2.90 times the long-run reference (10th to 90th percentile range 1.58 to 4.74; Fig. S2), and the Spearman correlation between them is 0.84 (95% interval 0.76 to 0.89). The correlation owes much to the wide range of mobilities in this subset: it is 0.93 among the 37 literature polymers but 0.68 among the 96 screening polymers. In the whole paired set the short

Table 1: Noise ceiling of short-run labels in the screening population: reliability of the mean of n runs, model-implied population correlation ceiling, and Gaussian-model recall of the highest expected short-run labels. Origin-averaged labels use lag $T/2$.

				Recall of the true top		
Label	Runs	Reliability	Ceiling	5%	10%	20%
End-point	1	0.44	0.66	36%	43%	54%
	2	0.61	0.78	46%	54%	63%
	3	0.70	0.84	54%	61%	68%
	4	0.76	0.87	58%	65%	72%
Origin-averaged	1	0.54	0.74	43%	50%	59%
	2	0.70	0.84	53%	61%	68%
	3	0.78	0.88	61%	67%	73%
	4	0.83	0.91	65%	70%	76%

run identifies 11 of the 13 best systems (Table 3), but these top sets consist almost entirely of literature polymers (13 of 13 and 24 of 27), which are easy to separate. Within the 96 screening polymers the short run still recovers 7 of the 10 best and 14 of the 19 best, against 10% and 20% by chance. These recalls are optimistic, because the short and long runs of a pair share their initial structure. A short run therefore provides useful coarse screening while uncertain differences limit fine ranking.

The median ratio of 2.90 is larger than the 1.95 obtained at the same lag from the origin-averaged long curves, and this slowdown explains the gap. The short runs behave like the first 5 ns of the long runs (median ratio 1.00), and that first segment is 1.42 times faster than the average of all segments, which is roughly what separates a single early window from a curve averaged over all time origins. Short labels in the dataset are therefore systematically biased upwards by the early, faster part of the trajectory, and more so for the screening polymers. For ranking, this bias matters only to the extent that it differs between systems.

3.5. Estimators and stopping within a run

Prefix estimates overstate the finite-lag reference: at 5 ns, typical errors are $\times 2.83$ for the end-point and $\times 2.02$ for a linear fit. Power-law extrapolation reduces error against $D_{\mathrm{app}}(25\,\mathrm{ns})$ to $\times 1.37$, but underestimates the late MSD slope (median ratio 0.54; Table S11). A subdiffusive power law does not approach normal diffusion; this is an empirical estimate of a specified

Table 2: Out-of-sample correlation of a random forest on monomer fingerprints with short-run labels (cross-validation split by system, or by chemical family of the monomer), against the noise ceiling of each label. The ceilings and their intervals rest on the run-to-run noise of the 94 replicated systems, not on the 6,003 systems of the first rows.

Evaluation target	Systems	Pearson r (95% CI)	Ceiling (95% CI)
One label, system split	6,003	0.69 (0.68 to 0.71)	0.66 (0.52 to 0.78)
Fine families (Tanimoto $\geq$ 0.6)	6,003	0.69 (0.67 to 0.70)	0.66 (0.52 to 0.78)
Coarse families (Tanimoto $\geq$ 0.4)	6,003	0.69 (0.68 to 0.70)	0.66 (0.52 to 0.78)
Replicas, first / second	94	0.64 / 0.68	0.66 (0.52 to 0.78)
Mean of two replicas	94	0.76 (0.63 to 0.84)	0.78 (0.65 to 0.87)

finite-lag quantity.

For the rules and reference tested, exponent- and precision-based stopping do not improve the fixed-length cost–error frontier (Figs. S3 and S4 and Table S12). Settings selected on half the systems can miss their target on the held-out half (Table S13). The shared reference data and conditional ion bootstrap limit this comparison. It does not rule out other stopping rules or establish that replication is preferable to extension for estimating equilibrium transport.

3.6. Spending the budget across candidates: adaptive allocation

The noise model measured above is compatible with the replicated systems: the differences between replicas show no departure from a Gaussian (excess kurtosis 0.13, Shapiro-Wilk $p = 0.81$). With it, how replicates are distributed matters more than how many are run (Fig. 4; values in Table S14). Uniform replicates improve slowly: for the top 10% of 6,003 candidates the recall rises from 45% with one run to 70% with four. Boundary-focused allocation reaches 72% with two runs per candidate on average and 82% with four, and matches four uniform replicates with 1.9 runs per candidate, a saving of $\times$2.1. The saving grows with the selectivity of the target: $\times$2.5 for the top 5% and $\times$3.1 for the top 1% (Table S15). Successive halving, which re-runs the current leaders in repeated rounds, performs similarly: with the end-point label it is better for the top 1% at small budgets (71% against 64% with 1.25 runs per candidate), and with the origin-averaged label it is worse at larger budgets (81% against 89% with four). Even a two-stage re-

Table 3: Recall of the best systems according to the 50 ns reference when they are selected from the 5 ns run of the same system (133 systems; the last rows only the 96 screening polymers). Intervals by bootstrap over systems; "Random" is the expected recall of a random selection of k systems. All recalls are optimistic, because the two runs of a pair share their initial structure.

Estimator, 5 ns run	Top	k	Hits	Recall (95% CI)	Random
End-point	5%	7	7	100% (57 to 100)	5%
	10%	13	11	85% (62 to 100)	10%
	20%	27	23	85% (70 to 93)	20%
Origin-averaged, lag $T/2$	5%	7	6	86% (57 to 100)	5%
	10%	13	10	77% (62 to 100)	10%
	20%	27	22	81% (70 to 93)	20%
End-point, screening only	5%	5	4	80% (40 to 100)	5%
	10%	10	7	70% (50 to 90)	10%
	20%	19	14	74% (47 to 89)	20%

screening, in which the budget left after the first run is spread evenly over a fixed shortlist of the best candidates, captures most of the gain: with a shortlist of the best $4k$ fixed in advance it saves $\times$n/a for the top 1%, $\times$2.3 for the top 5% and $\times$1.8 for the top 10%, and the best of three shortlist sizes in hindsight gives ($\times$3.2, $\times$2.3 and $\times$1.8). Most of the saving therefore comes from not spending replicates on candidates that the first run already places far from the cut-off; allocating them where the selection is still in doubt adds the rest, and does so without a shortlist size to tune. The price is sequential work: in batches of 5% of the candidates, matching four uniform replicates takes 18 additional rounds of simulation and analysis for the top 10% and 6 for the top 1%, against a single extra round for the two-stage re-screening. Where each round waits in a queue, the two-stage scheme is the practical alternative, but its saving depends on the shortlist size and should be evaluated with that choice fixed in advance. Nor does the saving depend on knowing the reliability exactly: planning with $r = 0.27$ or 0.61 instead of 0.44 gives savings of $\times$2.1 and $\times$2.1 for the top 10% (Table S16), and if the true reliability lies at either end of its interval the saving is $\times$1.6 to $\times$2.5 for the top 10% and $\times$2.7 to $\times$3.2 for the top 1%. The result holds with the empirical distribution of the short runs and with the origin-averaged label, and knowing that slow systems are noisier does not improve the allocation (Supplementary Material). For the expected short-run target and this

noise model, an initial run followed by selective new-structure replication is preferable to uniform replication at the tested budgets.

These gains are predictions of the noise model, since the public data contain at most two runs per system; the replicated systems allow only a partial check, which is consistent with them but has little statistical power (Table S17). The model's probabilities of exceeding the top-10% cut-off are informative on real runs, improving the Brier score of the base rate by 35% with a calibration slope of 1.00 (95% interval 0.57 to 2.26). In the real campaign on 92 systems, sending the 46 second runs to the boundary recovers 78% of the top 10% according to the long runs, as many as giving a second run to every system (78%), against 71% for randomly placed second runs and 67% with first runs only. With 9 systems in the top set the difference from random placement is not significant (95% interval from −12 to +16 percentage points), and for the top 20% no strategy improves on the first run alone. The long runs share their structure with the first runs, which probably favours the first run over any second run from a new structure, by an amount we cannot quantify. The calibration check, in turn, concerns a second noisy run rather than the latent value that the allocation targets. Noise parameters for the campaign are fitted to the available campaign pairs. The real runs therefore do not contradict the simulated gains, but they cannot show that adaptive allocation beats random placement.

3.7. Related transport labels

The anion diffusivity and the platform's cluster Nernst–Einstein conductivity extend the audit to related observables (Table 4). The latter combines end-point ion diffusivities and cluster populations [31]; we use the distributed labels in their reported units. This analysis does not establish the accuracy of those absolute units or validate the cluster approximation against collective charge displacement. For a matched ion comparison we use origin-averaged diffusivities at half the run length.

Anion labels are more reliable than lithium labels (0.79 against 0.54), and their short–long ranking agreement is higher (0.83 against 0.70 among screening systems). Correlations against replica means increase for both ions. The conductivity has reliability 0.63 (95% interval 0.51 to 0.74), with predictor correlation 0.73 and estimated ceiling 0.79. These point comparisons do not identify which chemical information the predictor lacks. Calibrated constant-noise campaigns again predict savings from adaptive replication, conditional on their expected-label targets.

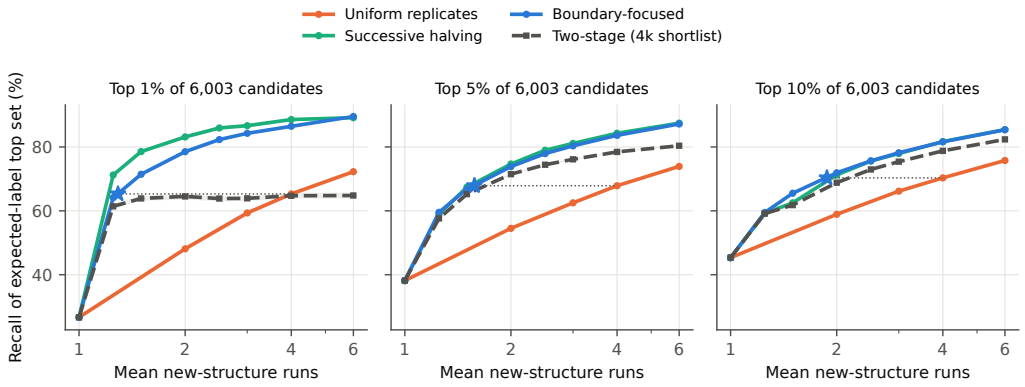


Figure 4: Recall of the true top set as a function of the mean number of 5 ns runs per candidate, for four ways of allocating the budget (200 simulated campaigns of 6,003 candidates with the measured noise; each run starts from a new, equilibrated structure). The true top set is the top k by the expected 5 ns label of each system, not by its long-time diffusivity. The two-stage shortlist is fixed at $4k$. Bands show 95% Monte Carlo intervals conditional on the calibrated model; markers show boundary budgets matching four uniform replicates.

Choosing the physical label matters. In screening runs, lithium diffusivity and conductivity have rank correlation 0.56 and their top deciles overlap by 47%, despite correlated replica differences (0.78). A conductivity screening campaign therefore needs a conductivity target. Agreement across these observables supports the audit within HTP-MD, but they share trajectories and physical information and are not independent tests of transfer to other regimes.

3.8. Practical guidance

The recommendations in Table 5 apply to the studied protocol, not to universal MD lengths. Before production, check energy drift and relevant structural and dynamical observables; stationary energy alone does not establish transport convergence. Report the estimator, lag and MSD regime. Prefer origin-averaged labels and distinguish within-run precision from accuracy against a long-time coefficient. Choose the physical target explicitly: lithium self-mobility and conductivity rank different candidates because correlated ion motion can carry little net charge [31].

Table 4: Results for the origin-averaged ion diffusivities at half the run length and the platform's cluster conductivity. The population is one original label per screening system; noise comes from 94 replica pairs. Allocation uses constant noise and targets expected short-run labels.

	Li^+	$TFSI^-$	Conductivity
Run-to-run noise (log)	0.28	0.19	0.31
Noise, slow / fast half	0.29 / 0.26	0.20 / 0.18	0.31 / 0.31
Reliability (95% CI)	0.54 (0.40 to 0.67)	0.79 (0.74 to 0.84)	0.63 (0.51 to 0.74)
Noise ceiling	0.74	0.89	0.79
Random forest vs one label	0.76	0.85	0.73
Random forest vs two-run mean (expected)	0.82 (0.84)	0.89 (0.90)	0.78 (0.77)
Spearman with 50 ns run, screening	0.70	0.83	0.77
Spearman with 50 ns run, literature	0.94	0.96	0.98
Runs matching 4 uniform replicates (top 10%)	1.8 (×2.3)	1.5 (×2.7)	1.7 (×2.4)

3.9. Limitations

All systems share one force field, one temperature and one salt; the examined transport labels are related observables and the numbers are specific to this setting, and the workflow, not the values, is what transfers to other campaigns. The long runs of the literature polymers were chosen by the dataset's authors; the screening systems with a replica (94) or a long run (96) show no detected difference in their marginal labels from the rest of the screening set, but they are few and largely the same systems (92 in common). The reference lag of 25 ns is not a converged limit and shares data with the truncated trajectories, so prefix-based errors at long T are optimistic. Self-diffusion is only a proxy for conductivity, and the cluster Nernst-Einstein conductivity neglects correlations between different clusters [31]; finite-size effects [32] are not corrected. The boxes are small (3.0 to 4.2 nm side, 1 to 15 chains), so each initial structure samples few chain conformations and packings; the structural part of the noise may be smaller in larger boxes, which would raise the reliability of a single run. The structural attribution additionally assumes independent dynamical histories; shared equilibration and system-specific batch effects cannot be excluded. We do not establish that the structural contribution persists at arbitrary run lengths. The noise model rests on 94 replicated pairs: the allocation gains are predictions of a model that the real runs, with 92 systems, do not contradict but cannot confirm; a campaign with several replicates per candidate would test them directly. No allocation result establishes the best policy for long-time transport when short-run biases change the ranking. The observables share trajectories and physical information, so their agreement is not independent cross-domain val-

Table 5: Decisions supported by this case study.

Objective	Action	Main limitation
Coarse screening	One short run; use an origin-averaged label	Similar candidates remain hard to separate
Select by expected short-run label	Replicate uncertain candidates; use a fixed shortlist if sequential rounds are costly	Savings are calibrated-model predictions, not validated savings for equilibrium transport
Compare similar candidates	Use independent initial structures	Replication reduces variance, not necessarily temporal bias
Evaluate a learned model	Split by system; use replicated test labels	The estimated population ceiling is uncertain and model-dependent
Estimate equilibrium transport	Extend and test relaxation and lag stability	Even 50 ns rarely establishes the diffusive regime here

idation. The small SCAYLE pilot checks protocol consistency; its annealed structures follow a different preparation and it cannot validate the allocation gains (Section S6). The analysis is largely exploratory: windows, thresholds, estimators and strategies were compared on the same data, and only the stopping-rule settings were chosen and evaluated on separate halves of the systems.

4. Conclusions

This audit separates two limitations of short MD labels: their mobility depends on observation lag and decreases with trajectory age, while independent initializations give substantial run-to-run variation. Replica noise accounts for about half the screening-label variance. The inferred prediction ceiling is uncertain; a fingerprint model is compatible with it, but model saturation is not established. The structural attribution is conditional on the assumed noise decomposition and does not prove irreducibility under longer simulation.

Calibrated campaign simulations predict substantial savings from selective replication when the objective is the expected short-run label. A single

re-screening round retains much of the gain; the available real runs cannot confirm superiority over random placement. These savings do not establish recovery of the best equilibrium transport properties. The related transport-label analyses support the usefulness of the audit within this dataset, while their numerical reliabilities differ. The transferable contribution is a procedure for specifying the label, measuring its noise and time dependence, and evaluating decisions against the intended physical target.

Data and code availability

The HTP-MD trajectories and curves are publicly available from the Toyota Research Institute platform [4]. The analysis scripts, the per-trajectory result tables and the scripts that generate every number, table and figure of this paper are available at `https://github.com/fhernandogallego/md-label-reliability` with the original analysis snapshot archived on Zenodo (`https://doi.org/10.5281/zenodo.23483460`); the revised scripts and derived inputs accompany this revision, and the raw trajectories are not redistributed. The analysis scripts were written with the assistance of Claude (Anthropic) and revised with Codex (OpenAI); the authors reviewed and verified the code. The scripts regenerate tables and figures from the released derived inputs. The SCAYLE pilot intervals are archived summary results; their per-run inputs are not included in the compact-input workflow, so those bootstrap intervals are not independently recomputed there.

Supplementary material

Supplementary Material associated with this article (data set details and validation of the trajectory processing; sensitivity of the regime census and of the lag dependence; run-to-run reproducibility; end-point estimator; paired runs; prefix-based estimators, stopping rules and held-out evaluation; within-run slowdown; exploratory ion aggregation; the new SCAYLE protocol pilot; robustness of the adaptive allocation and its check on real runs) is provided as a separate file.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Acknowledgements

The authors acknowledge SCAYLE (Supercomputación Castilla y León, León, Spain) for providing access to its computing infrastructure.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) and Codex (OpenAI) to assist with writing and editing the manuscript, as none of the authors is a native English speaker. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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